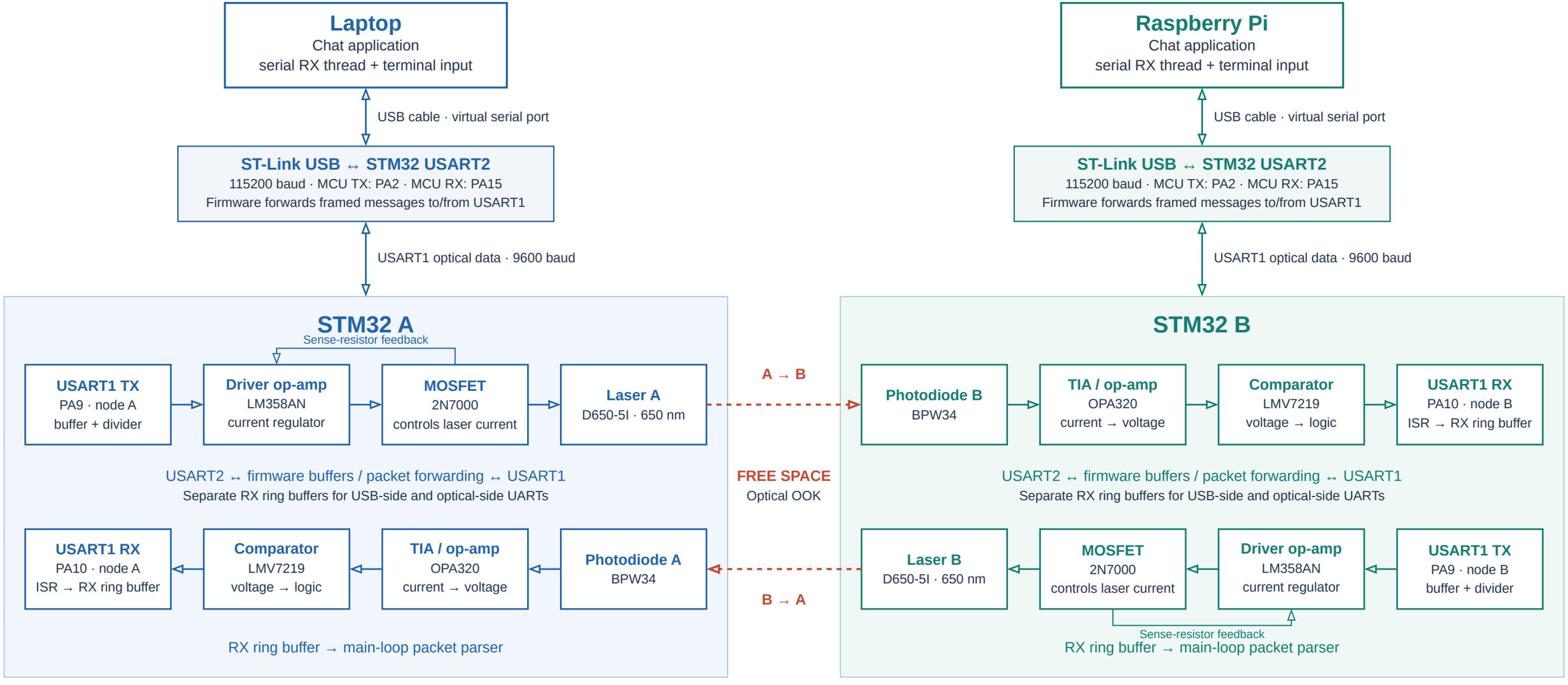
Laptop ↔ optical link ↔ Raspberry Pi

Chat applications on both hosts · two STM32F303K8 nodes · bidirectional optical communication



Solid arrows: electrical / data connections · Dashed arrows: light · Optical RX buffers feed packets back to the local host

TX buffer + divider: SN74AHCT125 and reference divider. Driver op-amp uses sense-resistor feedback to regulate laser current.

Functional signal flow, not a wiring schematic. Supplies, protection, biasing, and loop compensation are omitted.